

CS2422 Assembly Language & System Programming

October 16, 2003

Today's Topics

- IA-32 Memory
- Have some fun with emulators (Apple II and MAME).

Study Guide

- Section 2.2: IA-32 Processor
- Section 2.3: IA-32 Memory Management

Modes of Operation

- Protected mode
 - native mode (Windows, Linux)
 - Real-address mode
 - native MS-DOS
 - System management mode
 - power management, system security, diagnostics
- Virtual-8086 mode
 - hybrid of Protected
 - each program has its own 8086 computer

Addressable Memory

- Protected mode
 - 4 GB
 - 32-bit address
- Real-address and Virtual-8086 modes
 - 1 MB space
 - 20-bit address

Early Intel Microprocessors

- Intel 8080
 - 64K addressable RAM
 - 8-bit registers
 - CP/M operating system
 - S-100 BUS architecture
 - 8-inch floppy disks!
- Intel 8086/8088
 - IBM-PC Used 8088
 - 1 MB addressable RAM
 - 16-bit registers
 - 16-bit data bus (8-bit for 8088)
 - separate floating-point unit (8087)

The IBM-AT

- Intel 80286
 - 16 MB addressable RAM
 - Protected memory
 - several times faster than 8086
 - introduced IDE bus architecture
 - 80287 floating point unit

Intel IA-32 Family

- Intel386
 - 4 GB addressable RAM, 32-bit registers, paging (virtual memory)
- Intel486
 - instruction pipelining
- Pentium
 - superscalar, 32-bit address bus, 64-bit internal data path

IA-32 Memory Management

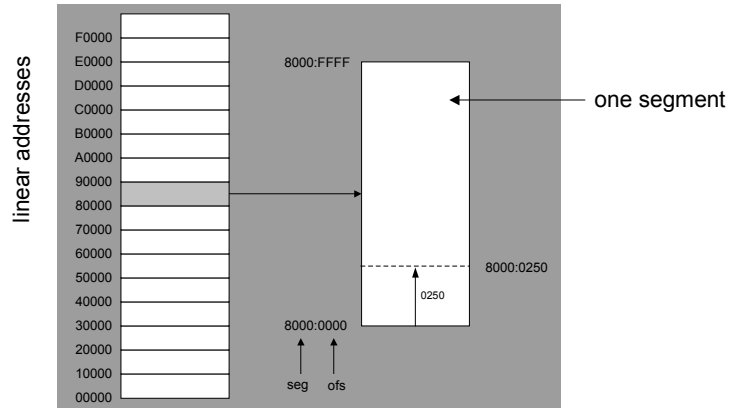
- Real-address mode
- Calculating linear addresses
- Protected mode
- Multi-segment model
- Paging

Real-Address mode

- 1 MB RAM maximum addressable
- Application programs can access any area of memory
- Single tasking
- Supported by MS-DOS operating system

Segmented Memory

Segmented memory addressing: absolute (linear) address is a combination of a 16-bit segment value added to a 16-bit offset



Calculating Linear Addresses

- Given a segment address, multiply it by 16 (add a hexadecimal zero), and add it to the offset
- Example: convert 08F1:0100 to a linear address

Adjusted Segment value:	0 8 F 1 0
Add the offset:	0 1 0 0
Linear address:	0 9 0 1 0

Protected Mode (1 of 2)

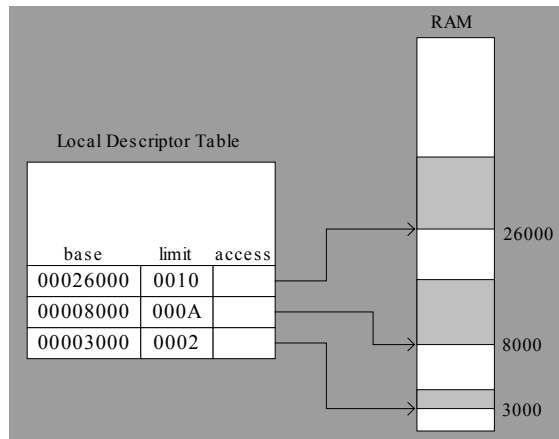
- 4 GB addressable RAM
 - (00000000 to FFFFFFFFh)
- Each program assigned a memory partition which is protected from other programs
- Designed for multitasking
- Supported by Linux & MS-Windows

Protected mode (2 of 2)

- Segment descriptor tables
- Program structure
 - code, data, and stack areas
 - CS, DS, SS segment descriptors
 - global descriptor table (GDT)
- MASM Programs use the Microsoft flat memory model

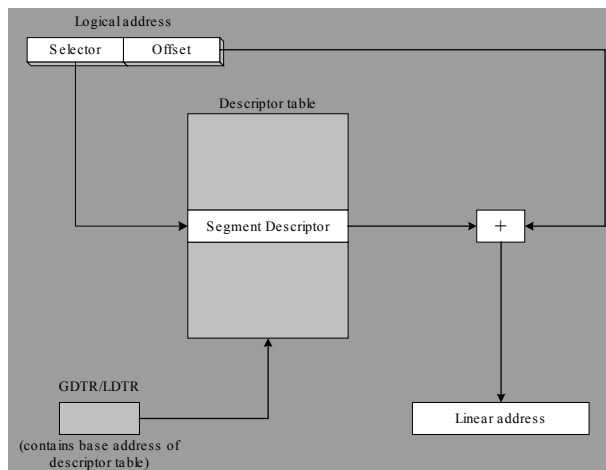
Multi-Segment Model

- Each program has a local descriptor table (LDT)
 - holds descriptor for each segment used by the program



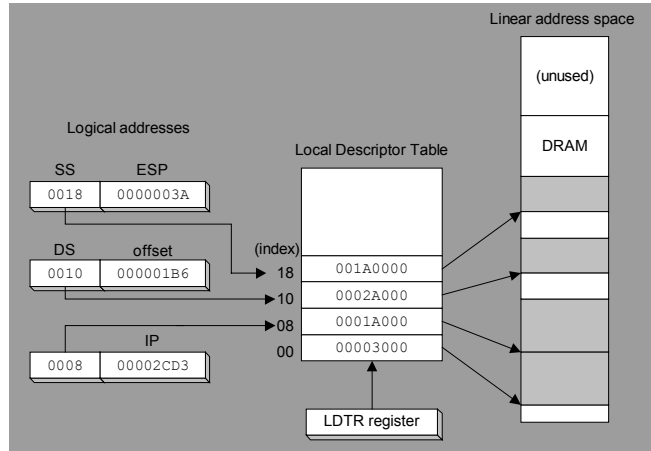
Converting Logical to Linear Address

The segment selector points to a segment descriptor, which contains the base address of a memory segment. The 32-bit offset from the logical address is added to the segment's base address, generating a 32-bit linear address.



Indexing into a Descriptor Table

Each segment descriptor indexes into the program's local descriptor table (LDT). Each table entry is mapped to a linear address:



Paging

- Supported directly by the CPU
- Divides each segment into 4096-byte blocks called pages
- Sum of all programs can be larger than physical memory
- Part of running program is in memory, part is on disk
- Virtual memory manager (VMM) – OS utility that manages the loading and unloading of pages
- Page fault – issued by CPU when a page must be loaded from disk

For More Information

- See Section 11.3